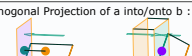
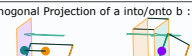


k-blades = Elements & Reflections = A ∨ B ∨ ... ∨ Z	k-reflections = Transformations = AB...Z
$\geq 1D$  Point(+reflection)	 Translation
$\geq 2D$  Line(+reflection)	 Translation
$\geq 3D$  Plane(+reflection)	 Translation

Geometric Gauges.	1DOF Translation Gauge
 Rotation	 Translation
 Point reflection	 Screw

Geometric Products - How they work.	Regressive Product avb → Select Identical
 Two Rotations Gauge 1 Gauge 2 Eliminate Identical	 Two Points Gauge 1 Gauge 2 Select Identical
 Point and Line Gauge Select Orthogonal	 Two lines Keep Intersection Make Orthogonal

Geometric Products - what they do.	Regressive Product avb → join
 Sandwich Product $aba^{-1} \rightarrow$ conjugate/action	 Regressive Product $avb \rightarrow$ join
 Point a, Line b	 Point a, Line b

Projective Geometric Algebra
Cheat Sheet by Steven De Keninck  $\mathbb{R}_{d,0,1}$ Euclidean $e_0^2 = 0$ $\mathbb{R}_{d+1}$ Elliptic $e_0^2 = 1$ $\mathbb{R}_{d,1}$ Hyperbolic $e_0^2 = -1$
GA Basics Orthogonal Basis Vectors $e_i \cdot e_j = \delta_{ij}$ Basis Blade $e_i \wedge e_j = e_{ij}$ Pseudoscalar $I = e_1 e_2 \dots e_d$ Dual basis blade x^* k -blade $e_1 \dots e_k$ k -reflection $e_1 \dots e_k$ Multivector $M = m_0 + m_1 e_1 + \dots + m_d e_1 \dots e_d$ Involute $\hat{x}$ Reverse $\hat{x}$ Dual x^* Norm $\ x\ $ and Ideal norm $\ x\ _\infty$

Geometric Products
Geometric Product between k -vector x and l -vector y $x \cdot y = \langle xy \rangle_{ k-l } + \langle xy \rangle_{ k-l +2} + \dots + \langle xy \rangle_{k+l}$ Inner Product $x \cdot y = \langle xy \rangle_{ k-l }$ Outer Product $x \wedge y = \langle xy \rangle_{k+l}$ Commutator Product $x \times y = \frac{1}{2}(xy - yx)$ Regressive Product $(x \vee y)^* = x^* \wedge y^*$ Sandwich Product $x \cdot y = \langle xy \rangle_{ k-l }$ Orthogonal projection of x onto l

$\mathbb{R}_{3,0,1}$ - 3D Euclidean PGA
Plane $ax + by + cz + d = 0$ Point (x, y, z) Line Plücker coordinates $a..f$ basis $1, e_1, e_2, e_3, e_{01}, e_{02}, e_{03}, e_{12}, e_{13}, e_{23}, e_{012}, e_{013}, e_{023}, e_{123}$ metric $+1, +1, +1, 0, 0, 0, 0, 0, 0, -1, -1, 0, 0, 0, 0, -1, 0$ dual $I, e_1, e_2, e_3, e_{01}, e_{02}, e_{03}, e_{12}, e_{13}, e_{23}, e_{012}, e_{013}, e_{023}, e_{123}$ reverse $1, e_1, e_2, e_3, e_{01}, e_{02}, e_{03}, e_{12}, e_{13}, e_{23}, e_{012}, e_{013}, e_{023}, e_{123}$ involute $1, e_1, e_2, e_3, e_{01}, e_{02}, e_{03}, e_{12}, e_{13}, e_{23}, e_{012}, e_{013}, e_{023}, e_{123}$ Join points v, w and line ℓ Meet planes p, q and line ℓ Distance point P to any x Angle between same grade x, y Distance between parallel x, y Common Normal Line ℓ_1, ℓ_2

Motors - General
Motor/Versor Squared Norm $M \bar{M} = s + tI$ Study inverse square root $\frac{1}{\sqrt{s + tI}} = \frac{1}{\sqrt{s}} - \frac{t}{2\sqrt{s}}I$ Normalize Motor/Versor $\bar{M} = \frac{M}{\sqrt{M \bar{M}}}$ Square Root of Motor $\sqrt{M} = 1 + \sqrt{M}$
Rotors - Group & Algebra
Logarithm of Motor $\log M = b(M)M^2 + c(M)I$ Exponential of Bivector $e^B = c + sB + tBI + msI$ Cayley Transform $B = \frac{1-M}{1+M}, M = \frac{1-B}{1+B}$

Rotors - Constructing/Finding
Rotor from Yaw, Pitch, Roll $R = e^{\frac{-\theta e_{12}}{2}} e^{\frac{-\phi e_{23}}{2}} e^{\frac{-\psi e_{31}}{2}}$ Rotate α around line ℓ $e^{\frac{-\alpha \ell}{2}}$ Translate δ along line ℓ $e^{\frac{\delta \ell}{2}} e_0$ Invariant Factorisation $M = TR = RT$ Euclidean Factorisation $M = TR_{origin}$ Rotor $R = e^{\frac{-\theta e_{12}}{2}} e^{\frac{-\phi e_{23}}{2}} e^{\frac{-\psi e_{31}}{2}}$ Motor k -blade y to k -blade x $\sqrt{\frac{x \cdot y}{y \cdot y}} \frac{x \pm y}{y}$ Direct motor points p_i to q_i $M_{i+1} = \sqrt{\frac{q_i \cdot \dots \cdot q_n}{q_i \cdot \dots \cdot q_n}} \frac{q_i \pm \dots \pm q_n}{q_i \cdot \dots \cdot q_n} \bar{M}_i$ Interpolating motor $R \rightarrow S$ $e^{t \log S \bar{R}}$ Average of $M_{1..n}$ from guess G_0 $G_{s+1} = \exp(\frac{1}{n} \sum_{j=1}^n \log(M_j \bar{G}_s)) G_s$

Kinematics & Dynamics
Forque = Force + Torque = line $f = d-2$ vector ℓ Velocity = lin + ang = hyperline $v = 2$ -vector ℓ Inertial duality x^* $x^* = x^*$ with coefficient scalar Euler's equations $\dot{M} = -\frac{1}{2} M \hat{v}, \hat{v} = (f - v^* \times v)^*$ Time derivative of point, line, plane x $\dot{x} = x \times v$ Kinetic Energy $e = \frac{1}{2} v^* \vee v$
Polygons and Meshes
Vertices v $v = (x e_1 + y e_2 + z e_3 + e_0)^*$ Edge Lines ℓ $\ell = v_i \vee v_j$ Face Planes f $f = v_i \vee v_j \vee v_k$ Mesh Surface Area $a = \frac{1}{2} \sum f_i $ Mesh Volume $v = \frac{1}{6} \sum f_i _\infty$

